# Python Gmail Email Program

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Time required: 30 minutes

Let's learn how to send email using Python and Gmail. Being able to send an email from Python opens up all sorts of possibilities.

## Setting up a Gmail Account for Email

If you don't have a Gmail account, you will want to create one.

Let's enable your Gmail account to receive connections from external programs, like Python.

1. Open your browser and access your Gmail account.
2. On the login screen → enter your Gmail username and password.
3. After the login → access the following URL: https://myaccount.google.com/signinoptions/two-step-verification
4. Enable the two-step verification on this account.

### Create an Application Password

1. You may also have to search around for the app passwords, it seems to be in different places in different accounts. Try these first.
   - https://security.google.com/settings/security/apppasswords
   - https://myaccount.google.com/apppasswords
2. At the same URL → **App passwords** → App name: **Python**

3. Take note of the randomly generated password.

You have finished the required steps for the Gmail integration.

## Tutorial 1 - Send Email with Python

Create a Python program named: **gmail_credentials.py**

```python
#!/usr/bin/env python3
"""
    Name: gmail_credentials.py
    Author:
    Created:
    Purpose: Credentials to send email through Python using Gmail
"""

SMTP_SERVER = "smtp.gmail.com"
# Secure SMTP port
PORT = 587

#---------- REPLACE WITH YOUR INFORMATION ------------#

LOGIN = "youremailaddress@gmail.com"
APP_PASSWORD = ""

#-----------------------------------------------------#
```

Create a Python program named: **send_gmail.py**

```python
#!/usr/bin/env python3
"""
    Name: send_gmail.py
    Author:
    Created:
    Purpose: Send email with Python and Gmail
"""
# Library to manage communcation with an SMTP server
import smtplib
# Library to create an email message
from email.message import EmailMessage
# Create secure context for tls encryption
import ssl
# Import gmail credentials and SMTP server settings
import gmail_credentials

#################################################################
#-------------- REPLACE WITH YOUR INFORMATION ------------------#
email_from = "William Loring <williamaloring@gmail.com>"
# A list containing one or more email addresses
email_dst = [
    "loringw@wncc.edu",
    "williamloring@hotmail.com"
]
content = """
This test message is sent from Python Bill.
"""
subject = "Email from Python Email Program"
#################################################################

#-------------------- CREATE EMAIL MESSAGE ---------------------#
message = EmailMessage()
message["From"] = email_from
message.set_content(content)
message["Subject"] = subject
message["To"] = email_dst
```

```python
#-------------------- CREATE EMAIL MESSAGE ---------------------#
message = EmailMessage()
message["From"] = email_from
message.set_content(content)
message["Subject"] = subject
message["To"] = email_dst

#------------------- SEND EMAIL MESSAGE -----------------------#
try:
    # Use with context manager to create an smtp_server object
    with smtplib.SMTP(
        gmail_credentials.SMTP_SERVER,
        gmail_credentials.PORT
    ) as smtp_server:
        # Show all communication with the server
        # This line can be commented out
        smtp_server.set_debuglevel(True)
        # Say enhanced hello to the smtp server
        smtp_server.ehlo()
        # Create a secure SSL context (connection object)
        context = ssl.create_default_context()
        # Request a TLS connection with the SMTP server
        smtp_server.starttls()
        # Login to the SMTP server
        smtp_server.login(
            gmail_credentials.LOGIN,
            gmail_credentials.PASSWORD
        )
        # Ask smtp_server to send out message
        smtp_server.send_message(
            message
        )
    print()
    print(25*"**")
    print("    Email message successfully sent.")
    print(25*"**")
    print()
except Exception as e:
    print(25*"**")
    print(f"Message not sent.")
    print(e)
    print(25*"**")
```

The example run contains all the chatter back and forth between your Python program and the mail server. The debug level is set to True so we can see how much background communication occurs sending a simple email message. The debug level line can be commented out.

Example run:

```
reply: retcode (250); Msg: b'2.1.0 OK ep25-20020a056870a99900b0011f390fdb0asm361
9396oab.12 - gsmtp'
send: 'rcpt TO:<loringw@wncc.edu>\r\n'
reply: b'250 2.1.5 OK ep25-20020a056870a99900b0011f390fdb0asm3619396oab.12 - gsm
tp\r\n'
reply: retcode (250); Msg: b'2.1.5 OK ep25-20020a056870a99900b0011f390fdb0asm361
9396oab.12 - gsmtp'
send: 'rcpt TO:<williamloring@hotmail.com>\r\n'
reply: b'250 2.1.5 OK ep25-20020a056870a99900b0011f390fdb0asm3619396oab.12 - gsm
tp\r\n'
reply: retcode (250); Msg: b'2.1.5 OK ep25-20020a056870a99900b0011f390fdb0asm361
9396oab.12 - gsmtp'
send: 'data\r\n'
reply: b'354  Go ahead ep25-20020a056870a99900b0011f390fdb0asm3619396oab.12 - gs
mtp\r\n'
reply: retcode (354); Msg: b'Go ahead ep25-20020a056870a99900b0011f390fdb0asm361
9396oab.12 - gsmtp'
data: (354, b'Go ahead ep25-20020a056870a99900b0011f390fdb0asm3619396oab.12 - gs
mtp')
send: b'From: William Loring <williamaloring@gmail.com>\r\nContent-Type: text/pl
ain; charset="utf-8"\r\nContent-Transfer-Encoding: 7bit\r\nMIME-Version: 1.0\r\n
Subject: Email from Python Email Program\r\nTo: loringw@wncc.edu, williamloring@
hotmail.com\r\n\r\n\r\nThis test message is sent from Python Bill.\r\n.\r\n'
reply: b'250 2.0.0 OK  1662293307 ep25-20020a056870a99900b0011f390fdb0asm3619396
oab.12 - gsmtp\r\n'
reply: retcode (250); Msg: b'2.0.0 OK  1662293307 ep25-20020a056870a99900b0011f3
90fdb0asm3619396oab.12 - gsmtp'
data: (250, b'2.0.0 OK  1662293307 ep25-20020a056870a99900b0011f390fdb0asm361939
6oab.12 - gsmtp')
send: 'QUIT\r\n'
reply: b'221 2.0.0 closing connection ep25-20020a056870a99900b0011f390fdb0asm361
9396oab.12 - gsmtp\r\n'
reply: retcode (221); Msg: b'2.0.0 closing connection ep25-20020a056870a99900b00
11f390fdb0asm3619396oab.12 - gsmtp'

**************************************************
    Email message successfully sent.
**************************************************
```

Test your program.

## Assignment 1 - Python CLI Email Program

Way back in prehistoric times, we used CLI (command line interface) email programs like pine. Everything was on Unix systems, not a GUI in sight.

Let's build our own blast from the past email program.

## Requirements

1. Add a creative program title

2. Add input for
    a. Destination email
    b. Subject
    c. Message
3. Allow user to send another message

Example run:

```
+-------------------------------------------------+
|            An Ancient Email Client              |
|       A CLI Email client from yesterday         |
+-------------------------------------------------+
Destination email address: loringw@wncc.edu
Subject: Another CLI Email Client
Enter message (One line only):
This is a one line message.

   Sending Email message . . .
   Email message successfully sent.

Send another email (Y/N): y
Destination email address: loringw@wncc.edu
Subject: A second test
Enter message (One line only):
This is the last message of the day.

   Sending Email message . . .
   Email message successfully sent.

Send another email (Y/N): n
```

Yes, you have successfully created a money-making app, a CLI email program in Python!

## Assignment Submission

- Send the instructor an email at loringw@wncc.edu
- Insert a screenshot of the program run
- Attach the completed program files
- Submit the assignment in Blackboard.